Convolution

1D (continuous, discrete):

$$f * g\ (x) = \int_{\alpha=-\infty}^{\infty} f(\alpha) g(x-\alpha) d\alpha$$

$$= \sum_{\alpha=0}^{N-1} f(\alpha) g(x-\alpha)$$

2D (continuous, discrete):

$$f * g\ (x,y) = \int_{\alpha=-\infty}^{\infty} \int_{\beta=-\infty}^{\infty} f(\alpha,\beta) g(x-\alpha, y-\beta) d\alpha d\beta$$

$$= \sum_{\alpha=0}^{N-1} \sum_{\beta=0}^{N-1} f(\alpha,\beta) g(x-\alpha, y-\beta)$$

Convolution Properties

- Commutative:
  f*g = g*f
- Associative:
  (f*g)*h = f*(g*h)
- Homogeneous:
  f*(λg)= λ f*g
- Additive (Distributive):
  f*(g+h)= f*g+f*h
- Shift-Invariant
  f*g(x-x0,y-yo)= (f*g) (x-x0,y-yo)

The Convolution Theorem

$$\Im\{f(x) * g(x)\} = \Im\{f(x)\}\ \Im\{g(x)\}$$

and similarly:

$$\Im\{f(x)g(x)\} = \Im\{f(x)\} * \Im\{g(x)\}$$